

Fundamental Assumptions of the Classical Linear Regression Model

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Introduction

The **Classical Linear Regression Model (CLRM)** is based on several key assumptions. These assumptions ensure that the **Ordinary Least Squares (OLS)** estimators are **BLUE** (Best Linear Unbiased Estimators), according to the Gauss-Markov theorem.

Consider the model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + \varepsilon_i,$$

where ε_i is the random error term.

1. Linearity in Parameters

The regression model is linear in the unknown parameters:

$$E(Y_i | X_i) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik}.$$

- The coefficients enter the model linearly.
- The predictors themselves do not have to be linear (e.g., X^2 , $\log(X)$ are allowed).

Example

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$

is linear in parameters and therefore satisfies this assumption.

2. Random Sampling (Independent Observations)

The sample consists of independently and randomly selected observations:

$$(Y_i, X_i), \quad i = 1, \dots, n.$$

This assumption ensures that information from one observation does not systematically affect another.

3. No Perfect Multicollinearity

No explanatory variable is an exact linear combination of other explanatory variables.

For example,

$$X_3 = 2X_1 + 5X_2$$

would violate this assumption.

Consequences

- OLS coefficients cannot be uniquely estimated.
- The matrix $X'X$ becomes singular.

4. Zero Conditional Mean (Exogeneity)

This is often considered the most important assumption:

$$E(\varepsilon_i | X_i) = 0.$$

This means the error term has no systematic relationship with the regressors.

Violations Arise From

- Omitted variables
- Measurement error
- Simultaneity (endogeneity)
- Sample selection bias

Consequence of Violation

OLS estimators become biased and inconsistent.

5. Homoscedasticity (Constant Variance)

The variance of the error term is constant across all observations:

$$\text{Var}(\varepsilon_i | X_i) = \sigma^2.$$

- Equal spread of residuals across fitted values indicates homoscedasticity.
- A funnel-shaped residual pattern indicates heteroscedasticity.

Consequences of Violation

- OLS coefficients remain unbiased.
- Standard errors become incorrect.
- Hypothesis tests and confidence intervals may be unreliable.

6. No Autocorrelation

Errors are uncorrelated across observations:

$$\text{Cov}(\varepsilon_i, \varepsilon_j) = 0, \quad i \neq j.$$

This assumption is especially important for time-series data.

Example of Violation

A positive residual today tends to be followed by a positive residual tomorrow.

Consequences of Violation

- OLS coefficients remain unbiased.
- Standard errors become inefficient and misleading.

7. Normality of Errors

$$\varepsilon_i \sim N(0, \sigma^2).$$

This assumption is not required for unbiasedness or for the Gauss-Markov theorem, but it is useful for:

- Exact t -tests
- Exact F -tests
- Confidence intervals in small samples

When the sample size is large, the Central Limit Theorem often makes this assumption less critical.

Summary Table

Assumption	Mathematical Form	Why Important
Linearity	$Y = X\beta + \varepsilon$	Correct model specification
Random Sampling	Independent observations	Representative sample
No Perfect Multicollinearity	No exact linear dependence among regressors	Unique estimation
Zero Conditional Mean	$E(\varepsilon X) = 0$	Unbiasedness
Homoscedasticity	$\text{Var}(\varepsilon X) = \sigma^2$	Efficient estimates and valid standard errors
No Autocorrelation	$\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$	Valid inference
Normality	$\varepsilon \sim N(0, \sigma^2)$	Exact small-sample tests

Gauss-Markov Assumptions

The first five assumptions:

1. Linearity in parameters
2. Random sampling

3. No perfect multicollinearity
4. Zero conditional mean
5. Homoscedasticity

constitute the core **Gauss-Markov assumptions**. Under these assumptions, OLS estimators are **BLUE** (Best Linear Unbiased Estimators).

Classical Normal Linear Regression Model

Adding the assumptions of

- No autocorrelation
- Normality of errors

yields the full **Classical Linear Regression Model (CLRM)** commonly presented in econometrics and advanced statistics courses.